Mohd Mafaiz Uddin

Data Scientist

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Education

Bachelor of Technology, *Jawaharlal Nehru Technological University.* 11/2021 – 06/2025
Computer Science Engineering and Data Science. Hyderabad, India

Data Science, *Innomatics Research Labs* 06/2025
Coursework in Python, Machine Learning, Deep Learning, Open CV, Exploratory Data Analysis. Hyderabad, India

Skills

PYTHON
- Python: Anaconda, Jupyter Notebook, syntax, Identifiers & Operators, Array, Numpy, Data Manipulation, Pandas.

Deep Learning
• ANN, TensorFlow, Keras, Dropout, Batch Normalization, Regularization, Early Stopping, Optuna.

Exploratory Data Analysis(EDA)
- Data Cleaning and Preprocessing, Handling Missing Values, Removing Duplicates, Data Visualization.

Machine Learning
• Regression, Classification, Feature Engineering, Feature Selection, Cross-Validation, Hyperparameter Tuning, Model Selection.

Open CV
• OpenCV, MediaPipe, Facial Landmarks, EAR, MAR, Real-Time Video Processing.

Projects

Swiggy Delivery Time Prediction | Machine Learning
- Developed a Machine Learning regression model to predict food delivery time using 45,502 Swiggy delivery records and features including distance, traffic, weather, rider, vehicle, city and order-time information.
- Performed data cleaning, exploratory data analysis (EDA), feature engineering and missing-value treatment to prepare structured delivery data for model training.
- Built Scikit-learn preprocessing pipelines using SimpleImputer, StandardScaler, OrdinalEncoder and OneHotEncoder for numerical and categorical features.
- Compared Linear Regression, Decision Tree, Random Forest and XGBoost models and applied RandomizedSearchCV with 5-fold cross-validation for hyperparameter tuning.
- Achieved 0.82 R^2 and 3.19-minute MAE on the test set, demonstrating strong delivery-time prediction performance.

Anti-Money Laundering Detection | Artificial Neural Network
- Developed an Artificial Neural Network (ANN) binary classification model to detect potentially illicit financial transactions using 72,770 transaction records with banking, account, currency, payment and transaction-value features.
- Performed data preprocessing, exploratory data analysis, categorical encoding and numerical scaling using OneHotEncoder and StandardScaler.
- Designed a TensorFlow/Keras neural network using dense layers, ReLU activation and sigmoid output for binary transaction classification.
- Applied class weighting, Dropout, Batch Normalization, L1/L2 regularization, EarlyStopping and Optuna hyperparameter optimization to address class imbalance and improve model performance.
- Achieved 94% test accuracy, with 30% recall and 33% F1-score for the laundering class, highlighting the challenge of detecting highly imbalanced fraudulent transactions.

AI-Powered Driver Drowsiness Detection | Computer Vision
- Developed a real-time Computer Vision-based Driver Drowsiness Detection System using Python, OpenCV and MediaPipe Face Landmarker for webcam-based facial landmark detection.
- Implemented Eye Aspect Ratio (EAR) for left and right eye monitoring to detect prolonged eye closure associated with driver fatigue.
- Implemented Mouth Aspect Ratio (MAR) to identify yawning and combined facial indicators for drowsiness detection.
- Developed real-time video processing, frame-based thresholding and audio alert mechanisms to provide immediate warnings when prolonged drowsiness is detected.
- Achieved real-time drowsiness detection with automated audio alerts; the project notebook does not report a validated numerical accuracy score, so no unsupported accuracy percentage is claimed.

T-Shirts Trends Analysis using Exploratory Data Analysis (EDA)
- Performed Exploratory Data Analysis (EDA) on a dataset containing 5,000+ T-shirt product records to identify trends in pricing, ratings, reviews, and product categories.
- Used Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to clean, process, and visualize the dataset for better analytical insights.
- Analyzed key variables including brand, price range, color, ratings, and customer reviews to understand customer preferences and market demand.
- Built visualizations such as bar charts, histograms, correlation heatmaps, and distribution plots to identify relationships between product ratings, pricing strategies, and sales popularity.
- Conducted data preprocessing by handling missing values, removing duplicates, and standardizing product attributes to achieve 90%+ data quality and accuracy.

Certificates

- Certification in Python Programming Innomatics Research Labs
- Certification in Applied Machine Learning Innomatics Research Labs
- Certification in Exploratory Data Analysis Innomatics Research labs

Languages

- English
- Hindi
- Telugu